

JANIS BERZINS

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PROFESSIONAL SUMMARY

Results-driven automotive engineering professional with over 15 years of experience in production, performance testing, and process optimisation. Proficient in vehicle testing protocols, system-level diagnostics, and implementing design improvements to enhance manufacturing efficiency. Adept at collaborating with cross-functional engineering teams to deliver high-quality solutions under stringent timelines. Since 2026, has applied this engineering discipline to founding and independently building two software products end-to-end — from strategy design through full-stack development and cloud deployment — combining deep technical expertise with hands-on software development capability.

PROFESSIONAL EXPERIENCE

Thin Ice Digital UG (haftungsbeschränkt) — Berlin, Germany

Founder | 2026 – Present

Runnr — Trade-Discipline Platform

- Designed and built a mobile-first Progressive Web App from the ground up (HTML, CSS, JavaScript), including state management, offline functionality, and CSV data import/export.
- Developed and deployed a Python-based automated trading bot (Alpaca Markets API integration, APScheduler task scheduling, SQLite logging) to a live cloud production environment (Railway).

Glacifraga Trading — Algorithmic Trading Engine

- Designed, coded, and backtested systematic trading strategies (moving-average crossover, RSI, ATR, and ADX-based signal logic) across multi-year historical market data, including stress-testing against major historical market dislocations.
- Built bottom-up, multi-year financial forecasting models (Python/Excel) and institutional-facing materials to support funding applications and partnership discussions.

Tesla Giga Berlin — Germany

Lead — Body Sealing & Water Management | June 2022 – December 2025

Water Management System Development

- Engineered water ingress prevention systems, integrating sealing technologies and structural reinforcements.
- Defined control plans to standardise quality assurance protocols, ensuring robust system performance.
- Drove continuous design iterations and refinements, optimising sealing integrity.

Global Body Leakage Initiative

- Co-developed a scalable body leakage mitigation project, addressing gaps in vehicle chassis and enclosure systems.
- Solution shared globally across Tesla's production facilities.
- Led ongoing refinement of sealing methodologies, enhancing NVH performance and durability.

Failure Mode Work Instructions

- Developed comprehensive FMA guidelines for service centres targeting water-related failure modes.
- Standardised diagnostic and repair workflows, reducing labour hours and operational costs.
- Integrated RCA principles to minimise reoccurrence of water-related failures.

New Program Contributions

- Collaborated on next-generation vehicle programs, focusing on NVH optimisation for acoustic and structural dynamics.
- Conducted design reviews and simulations to mitigate leak, noise and vibration.

Wade Test Execution

- Led wade testing protocols at IDIADA to validate vehicle sealing and structural integrity under water immersion scenarios.
- Developed test plans aligned with ISO and SAE standards, ensuring compliance with environmental durability requirements.
- Analysed test data to refine component designs and sealing interfaces.

Component Validation Test Plans

- Designed and executed validation test plans for new and upgraded vehicle components, focusing on water management metrics.
- Ensured compliance with Tesla's performance specifications through iterative testing and design feedback.

Cross-Functional Network Development

- Established a robust interface with Berlin Gigafactory production teams, streamlining NVH issue resolution in manufacturing workflows.

- Maintained technical alignment with vehicle engineering and global NVH counterparts, facilitating knowledge transfer and standardised practices.
- Utilised JIRA and Confluence for cross-team project tracking, issue management and technical documentation workflows.

Rolls-Royce Motor Cars — Chichester, UK

Mechanical Engineering Technician | Feb 2018 – March 2021

- Performed dynamic on-road and track performance evaluations utilising diagnostic tools to identify and rectify manufacturing and design defects, ensuring compliance with OEM specifications.
- Collaborated with design and development engineers to integrate final-stage design modifications, optimising vehicle performance and aligning with manufacturer and dealer technical standards.
- Ensured adherence to ISO 9001 quality standards, health and safety regulations, and industry-specific protocols during all testing and production phases.
- Analysed production workflows using lean manufacturing principles.

BMW Mini Plant Oxford — Oxford, UK

Mechanical Engineering Technician | Nov 2011 – March 2021

- Designed and developed technical solutions for prototype vehicle builds, focusing on system integration and performance validation during new product development cycles.
- Conducted root cause analysis on production line systems, implementing corrective actions that reduced defect rates by 10% through enhanced quality control measures.
- Mentored junior technicians in advanced diagnostic techniques and production methodologies, fostering a culture of technical excellence and continuous improvement.
- Maintained strict compliance with supplier and manufacturer technical specifications, ensuring all met predefined engineering tolerances and standards.

EDUCATION

- RISEBA University of Applied Sciences, Riga — Bachelor of Business Management, Logistics & Marketing | Sept 2002 – Dec 2006
- Valmiera Pārgauja Gymnasium, Valmiera — Certificate of General Secondary Education | Sept 1993 – June 2002

KEY SKILLS

- **Technical Expertise:** Vehicle dynamics testing, root cause analysis, system integration, lean manufacturing, quality assurance (ISO 9001).
- **Software Development:** Python (pandas, NumPy), REST API integration (Alpaca Markets), SQLite, FastAPI, cloud deployment & DevOps (Railway), front-end development (HTML, CSS, JavaScript — Progressive Web Apps), Git, Excel/financial modelling (openpyxl), task automation & scheduling (APScheduler).
- **Software Proficiency:** Microsoft Office Suite, iOS, JIRA, Confluence, Tesla-specific diagnostic tools.
- **Languages:** Fluent English & Latvian; conversational German and Russian.
- **Engineering Knowledge:** Mechanical system design, failure mode analysis, production line optimisation, health & safety compliance.

PROFESSIONAL ATTRIBUTES

- Analytical mindset with a systematic approach to troubleshooting and process optimisation.
- Strong interpersonal and technical communication skills; adept at collaborating with multidisciplinary engineering teams.
- Meticulous attention to detail, ensuring precision in technical documentation and execution.
- Resilient under pressure, consistently delivering results in fast-paced environments.

REFERENCES

Hannes Swart — Senior NVH Engineer, Tesla EMEA Vehicle Engineering | dswart@tesla.com

Leonard Haeck — Manager, Tesla EMEA Service Development | LHaeck@tesla.com

Nicholaus Owen — Staff Body Repair Technical Specialist, Tesla Europe Service Business Operations | nicowen@tesla.com

Alan Auld — Showbay Special Vehicle Operations, BMW Mini Plant Oxford | alan.auld@bmwgroup.com